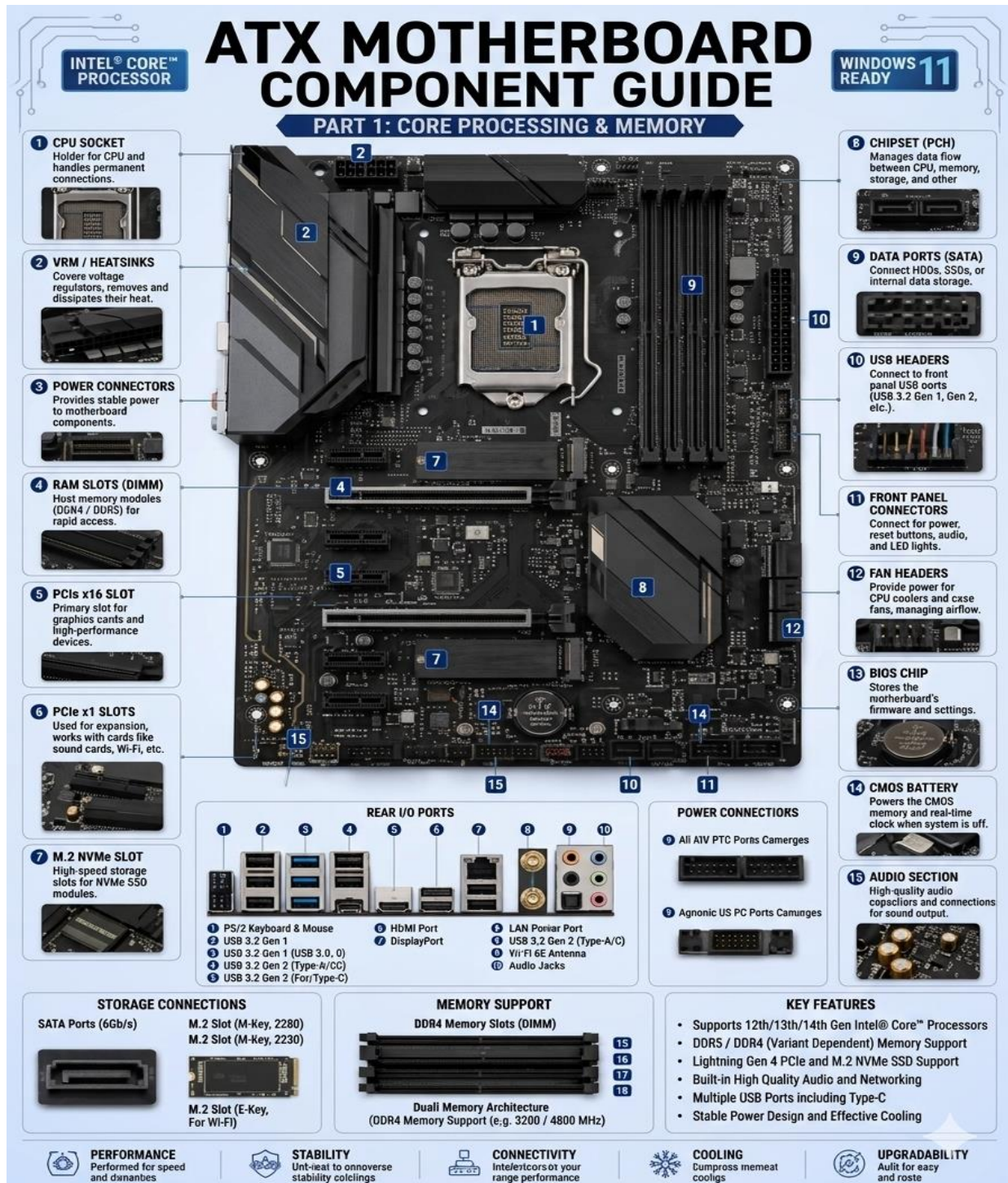


Day 2

Tasks

1. Draw a labelled diagram of an ATX motherboard and mark at least 8 different slots and ports (such as RAM slots, PCIe slots, SATA ports, USB headers, CPU socket, power connector, etc.).



Day 2

2. Research and list 3 popular CPU socket types (for example: LGA1151, AM4, LGA1200) and name one compatible CPU model for each.

❖ **Popular CPU Socket Types: -**

A CPU socket is the physical connector on a motherboard where the processor is installed. The CPU and motherboard must use a compatible socket type.

CPU Socket	Manufacturer	Example Compatible CPU	Typical Generation
LGA1151	Intel	Intel Core i5-9600K	8th & 9th Gen Intel
AM4	AMD	AMD Ryzen 5 3600	Ryzen 1000–5000 series*
LGA1200	Intel	Intel Core i5-10400	10th & 11th Gen Intel

The three-example CPU/socket combinations are documented as FCLGA1151-2 + i5-9600K, AM4 + Ryzen 5 3600, and FCLGA1200 + i5-10400.

1. LGA1151: -

LGA1151 was used by several Intel desktop generations, including 8th- and 9th-generation processors. One compatible processor is the Intel Core i5-9600K.

2. AM4

AM4 is an AMD desktop CPU socket. A popular processor using this socket is the AMD Ryzen 5 3600.

3. LGA1200

LGA1200 is an Intel desktop socket used by 10th- and 11th-generation processors. One example is the Intel Core i5-10400.

Important: A CPU cannot simply be installed in any motherboard. The socket must match, and the motherboard's chipset/BIOS also needs to support that particular processor. For example, Intel notes that LGA1151 and LGA1200 processors are electrically incompatible.

Day 2

3.Explain in your own words the primary functions of the CPU and the chipset on a motherboard, and describe how they work together when you open a music app like Spotify.

❖ CPU and Chipset Functions

1. CPU (Central Processing Unit)

The CPU is the main processor of the computer. It executes instructions, performs calculations, and processes the data required by software. When you open an application such as Spotify, the CPU runs the instructions that launch the program and manages its operations.

2. Chipset

The chipset acts as a communication controller between the CPU and many other components on the motherboard. It helps manage communication with devices such as storage, USB devices, expansion slots, and other motherboard-connected hardware.

How They Work Together When Opening Spotify

When you click the Spotify icon:

1. You click Spotify → The operating system receives your command.
2. CPU processes the request → The CPU executes the instructions needed to start Spotify.
3. Data is retrieved → Spotify's program files are loaded from storage into RAM, with the CPU coordinating the processing.
4. Chipset manages communication → The chipset/platform controllers help coordinate communication between the CPU and connected hardware such as storage, USB devices, and other I/O components.
5. CPU runs Spotify → The CPU processes Spotify's instructions and handles tasks such as displaying the interface and responding to your clicks.
6. Music plays → Audio data is processed and sent through the computer's audio hardware to your headphones or speakers.

In simple words: The CPU does the actual processing, while the chipset helps coordinate communication between the CPU and various motherboard-connected components. Together, they allow an application like Spotify to load, respond to your commands, and play music smoothly.

Day 2

4. Find a clear photo of a real motherboard (from the internet or your own device), and identify at least 5 different ports or slots by circling them and labelling each one.

